



Multimodal Taylor Series Network for Misinformation Detection

Jiahao Sun
College of Computer Science, MoE
Key Lab of DISSec
Nankai University
Tianjin, China
sunplus@mail.nankai.edu.cn

Chen Chen
College of Computer Science, MoE
Key Lab of DISSec
Nankai University
Tianjin, China
nkchenchen@nankai.edu.cn

Chunyan Hou
School of CSE
Tianjin University of Technology
Tianjin, China
chunyanhou@163.com

Yike Wu*
School of Journalism and
Communication, Lab of Intelligent
Publishing Technology and Standards
Nankai University
Tianjin, China
wuyike@nankai.edu.cn

Xiaojie Yuan
College of Computer Science, MoE
Key Lab of DISSec
Nankai University
Tianjin, China
yuanxj@nankai.edu.cn

Abstract

With the rapid development of the Internet and the widespread use of social media, the proliferation of multimodal misinformation combining images and text poses serious risks to societal trust, individual well-being, and the integrity of AI models trained on such data. Recently, the automatic detection multimodal misinformation has become an essential area of research. However, traditional methods often rely on hierarchical neural networks that compress and fuse modalities, potentially overlooking deeper interactions between modalities and reducing model interpretability. In this paper, we present a novel Multimodal Taylor Series (MTS) network for detecting multimodal misinformation. The MTS network leverages Taylor series expansion to explicitly capture both low-order and high-order interactions between modalities, which also enhances interpretability by decomposing the model's processing into distinct terms. Additionally, the proposed MTS network avoids exponential parameter growth and maintains linear scalability, allowing the model to effectively capture complex cross-modal correlations. Extensive experiments on three benchmark datasets demonstrate that the MTS network significantly outperforms state-of-the-art models. We have open-sourced the code and logs at: <https://github.com/OneForAllSama/MTS>.

CCS Concepts

- **Information systems** → **Multimedia information systems**; •
- Theory of computation** → **Design and analysis of algorithms**;
- **Computing methodologies** → *Machine learning*.

*Corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

WWW '25, April 28-May 2, 2025, Sydney, NSW, Australia

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-1274-6/25/04
<https://doi.org/10.1145/3696410.3714719>

Keywords

Multimodal, Misinformation Detection, Taylor Series Network

ACM Reference Format:

Jiahao Sun, Chen Chen, Chunyan Hou, Yike Wu, and Xiaojie Yuan. 2025. Multimodal Taylor Series Network for Misinformation Detection. In *Proceedings of the ACM Web Conference 2025 (WWW '25)*, April 28-May 2, 2025, Sydney, NSW, Australia. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3696410.3714719>

1 Introduction

With the rapid growth of the internet and social media, the proliferation of multimodal content combining images and text has significantly facilitated the spread of misinformation, posing substantial risks to both societal stability and individual well-being [14, 30]. In the era of large language models, the use of harmful information such as misinformation in training data can result in biased and erroneous outputs, including hallucinations [11], severely misleading users. Consequently, the detection of multimodal misinformation have become critically important.

Traditional approaches [4, 15, 19, 33] to multimodal misinformation detection typically employ hierarchical neural networks that progressively compress multimodal data. These methods often process image and text features separately, followed by modality fusion through simple feature concatenation or attention mechanisms. However, they frequently struggle to effectively capture the intricate, deep interactions between modalities [1]. Furthermore, the data flow in such methods is implicit, making it unclear which specific parameters are responsible for capturing relevant features, thus limiting model interpretability.

To address these limitations, we propose the Multimodal Taylor Series (MTS) network, which effectively handles the inherent complexity of multimodal data. Our approach treats image and text modalities as two variables and leverages a bivariate Taylor series expansion to approximate the mapping from network input to output. As shown in Figure 1, using the sine function as an example, the approximation becomes increasingly accurate as the order of the Taylor series increases, capturing finer details of the function.

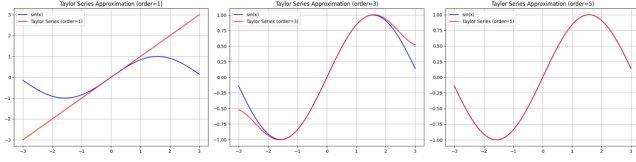


Figure 1: An illustration of the impact of gradually increasing the order of the Taylor series on function approximation accuracy.

Utilizing a bivariate Taylor series expansion for multimodal misinformation detection offers two key advantages. First, the Taylor series, with its different orders, can explicitly model both shallow and deep interactions between image and text modalities, from low-order to high-order interactions. Second, it decomposes the data flow into distinct terms, such as modality-independent terms and cross-modal interaction terms, enabling clear identification of the roles of different parameters and thereby enhancing model interpretability.

To prevent the exponential growth of parameters typically associated with increasing the order of the Taylor series, we simplify the expansion by applying modality-wide partial derivations of the encoded features and generating expansion terms in an efficient way. This optimization improves both the scalability and practicality of the model, making it easy to implement and deploy in real-world applications. In the experiments, we evaluate the proposed method on benchmark datasets for fake news and sarcasm detection. The results consistently show that the MTS network outperforms state-of-the-art methods. Additionally, we also provide a detailed analysis demonstrating the interpretability of the model.

2 Related Work

2.1 Misinformation Detection

With the widespread use of the Internet and social media, multimodal misinformation detection has recently emerged as an important research area. For instance, CAFE [4] combines unimodal features with cross-modal correlations by applying cross-modal fuzzy learning and modal fusion. FND-CLIP [33] employs CLIP to perform normalized and weighted fusion, reducing redundant information in both image and text features. LogicDM [19] leverages fine-grained word-level embeddings processed by an LSTM, incorporating additional predicate features to enhance interpretability and performance. BMR [29] adapts image and text features using consistency learning, followed by feature extraction and fusion through a multi-expert network. FSRU [15] improves rumor detection by integrating spectral features from both modalities. These methods are typically based on deep neural networks.

In contrast, this paper introduces a novel network architecture based on Taylor series expansion, which effectively captures interactions between modalities at various orders while offering clearer model interpretability.

2.2 Polynomial Networks

Research has demonstrated that polynomial methods can achieve superior fitting with fewer parameters, even without activation

functions [5, 8, 20]. For example, Π -Nets [5], a new class of deep convolutional neural networks (DCNNs), utilize polynomial neural networks for function approximation. These networks employ unique jump connections to form higher-order polynomials, extending traditional compositional paradigms. Similarly, CAT [8] uses a concept encoder alongside a polynomial network based on Taylor expansions, delivering strong performance across various single-modal tasks.

However, most prior work has focused on single-modal scenarios, limiting its applicability to multimodal tasks. One major limitation of previous methods is the exponential growth in the number of parameters with increasing polynomial degree, making it impractical to model higher-order complex functions. Polynomial structures of only 2-3 degrees struggle to capture intricate distributions of multimodal data and deep interactions between modalities. In contrast, our method is specifically designed for multimodal scenarios, ensuring that parameter growth remains linear with increasing polynomial degree. This addresses the limitations of earlier methods, which suffer from reduced fitting capacity due to exponential parameter growth.

3 Methodology

3.1 Overview

The overall framework is depicted in Figure 2. Given a multimodal input sample, we first encode the image and text modalities into their respective feature representations, e_i for the image and e_t for the text. For the image modality, we utilize ResNet followed by a linear layer to extract the encoded feature. For the text modality, we input the text into BERT, and extract the CLS token, which is then passed through a linear layer to obtain the encoded text feature. These encoded features are then fed into the proposed Multimodal Taylor Series (MTS) network to generate the final feature representations, denoted as M_n and N_n . Subsequently, we perform an element-wise addition of the two final feature representations, and pass the result through a linear layer to produce the logits for final classification.

The MTS network is derived from the Taylor series expansion and consists of multiple layers. As the number of layers in the network increases, the overall function represented by the network becomes more expressive, capturing increasingly complex interactions between the image and text modalities. In the next sections, we elaborate how the MTS network is derived from the original Taylor series expansion and adapted for practical implementation. To maintain mathematical clarity, henceforth we use $\vec{x}$ to denote the image feature e_i , and $\vec{y}$ to denote the text feature e_t .

3.2 Multimodal Taylor Series Network

In mathematics, Taylor's theorem states that a polynomial function of d variables can be approximated using a Taylor series expansion, as shown below:

$$f(\vec{x}) \approx \sum_{k=0}^N \frac{1}{k!} \left[\sum_{i=1}^d \left(\Delta x_i \frac{\partial}{\partial x_i} \right) \right]^k f|_{x_0}, \quad (1)$$

where $\vec{x} \in \mathbb{R}^d$, and N denotes the order of the Taylor series. To adapt this for multimodal data, we extend the equation to a

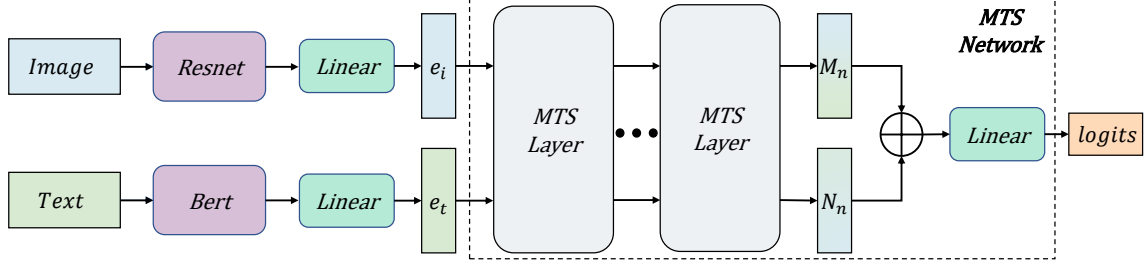


Figure 2: The overview of the proposed framework. Note that all the linear layers are single-layer linear transformations without activation functions, in order to preserve the original structure and properties of the Taylor series expansion.

bivariate form to better represent the interactions across multiple modalities. For simplicity, we perform the Taylor series expansion at the origin:

$$f(\vec{x}, \vec{y}) \approx f(\vec{0}, \vec{0}) + \sum_{k=1}^N \frac{1}{k!} \left(\sum_{i=1}^d x_i \frac{\partial}{\partial x_i} + \sum_{j=1}^d y_j \frac{\partial}{\partial y_j} \right)^k f \Big|_{(\vec{0}, \vec{0})}. \quad (2)$$

When incorporating tensors, this can be written as:

$$f(\vec{x}, \vec{y}) \approx f(\vec{0}, \vec{0}) + \sum_{k=1}^N \left(W^{[k]} \prod_{i=2}^{k+1} \hat{\times}_i [\vec{x}, \vec{y}]^\top \right), \quad (3)$$

where $\vec{x} \in \mathbb{R}^d$ and $\vec{y} \in \mathbb{R}^d$ are the feature representations of the image and text modalities, respectively, and $[\vec{x}, \vec{y}] \in \mathbb{R}^{2d}$ denotes the concatenation of the two features. $W^{[k]} \in \mathbb{R}^{\prod_{i=2}^k \times (2d)}$ represents the k -th order derivative parameters of the function, and $\hat{\times}_j$ denotes the mode- n matrix product [13]. The term $\prod_{i=2}^{k+1} \hat{\times}_i$ represents the operation of applying the k -th order derivative of the parameter $W^{[k]}$ and performing element-wise multiplication with $[\vec{x}, \vec{y}]$. The number of parameters required to fit the coefficients $W^{[k]}$ in the N -th order polynomial is:

$$O(f(\vec{x}, \vec{y})) = \sum_{k=1}^N (2d)^k = \frac{2d \times (1 - (2d)^N)}{1 - 2d}, \quad (4)$$

$$O(f(\vec{x}, \vec{y})) = (2d)^N. \quad (5)$$

As shown in Eq. 5, the number of parameters grows exponentially with the order N . While this approach is feasible when N is small, it becomes impractical for larger values of N , as it requires an excessive number of parameters to capture the complex interactions among different modalities. To address this scalability issue, we propose reducing the parameter size in Eq. 3.

3.3 Modality-Wide Partial Derivation of Encoded Features

To mitigate the exponential growth of parameters in Eq. 3 with increasing order N , we treat each modality's encoded feature as a whole for partial derivation, rather than treating each dimension of the feature as a separate variable. For example, for the image feature $\vec{x} \in \mathbb{R}^d$, we compute $\frac{\partial}{\partial \vec{x}}$ instead of $\left\{ \frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2}, \dots, \frac{\partial}{\partial x_d} \right\}$. As

a result, Eq. 3 transforms into:

$$f(\vec{x}, \vec{y}) \approx \sum_{k=0}^N \left(W_{\vec{x}}^k \vec{x}^\top + W_{\vec{y}}^k \vec{y}^\top \right)^k, \quad (6)$$

where the Hessian matrices $W_{\vec{x}}^k$ and $W_{\vec{y}}^k \in \mathbb{R}^{d \times d}$ represent the k -th order partial derivative matrix. The number of parameters required to fit this improved expression is:

$$O(f(\vec{x}, \vec{y})) = \sum_{k=1}^N (2d)^2 = N \times (2d)^2. \quad (7)$$

This reduces the parameter count from $(2d)^N$ to $N \times (2d)^2$, transforming the parameter growth from exponential to linear and significantly improving scalability. As a result, the model is capable of fitting higher-order, more complex functions.

Moreover, treating the modality-wide encoded feature as a whole is well-founded in multimodal learning. The semantic integrity of an encoded image or text feature should be preserved in the form of a complete vector, which is crucial for modality representation learning. If we treat each dimension separately, we risk breaking the semantic integrity of the feature and ignoring the interconnections between different dimensions, thereby weakening the expressive power of the encoded features.

3.4 Efficient Generation of Taylor Series Expansion Terms

Although Eq. 6 reduces the number of parameters, directly implementing the network architecture based on this equation remains suboptimal. In Eq. 6, the features of the two modalities are simply linearly added, meaning they are immediately combined into a single representation without preserving the distinct information from each modality. However, in multimodal learning, due to the semantic and modality gaps present in the encoded features of different modalities [17], such an early combination can negatively impact the representation learning of modality-specific features and reduce the model's ability to capture cross-modal interactions effectively.

To address this, one straightforward solution is to transform Eq. 6 into its full expansion form. The expanded expression must include terms of all orders from 0 to N , covering both single-modality features and cross-modal combination terms. Omitting coefficient

matrices for clarity, the cumulative sum of all terms can be expressed as follows:

$$A^N = \sum_{k=0}^N \sum_{i=0}^k (x^i \times y^{k-i}), \quad (8)$$

The relationship between the total number of terms and the order N is given by:

$$O_{\text{term}}(f(\vec{x}, \vec{y})) \approx 2^N. \quad (9)$$

As the order N increases, the number of terms grows exponentially, resulting in a proportional increase in the number of parameters. Consequently, implementing the full expansion of Eq. 6 is impractical. Manually setting all single-modality and cross-modal terms would be cumbersome, and the exponential growth in parameters would severely limit the model's scalability.

To solve this issue, we transform the process of generating all terms in the full expansion of Eq. 6 into a simpler and more efficient process for practical implementation. Specifically, we define two bivariate functions, $\mathcal{M}(\cdot, \cdot)$ and $\mathcal{N}(\cdot, \cdot)$, as follows:

$$\mathcal{M}(a, b) = a + ab + a^2, \quad \mathcal{N}(a, b) = b + ab + b^2, \quad (10)$$

where a and b are two variables. We further define the initial terms M_0 and N_0 as:

$$M_0 = \mathcal{M}(\vec{x}, \vec{y}), \quad N_0 = \mathcal{N}(\vec{x}, \vec{y}). \quad (11)$$

We can divide M_0 (and similarly N_0) into three components for finer analysis. The term $\vec{x}$ serves as a linear term that preserves low-order features. The term $\vec{x}\vec{y}$ acts as an interaction term, capturing mutual information between modalities and elevating $\vec{x}$ and $\vec{y}$ to higher-order terms. Lastly, $\vec{x}^2$ functions as a squared term, focusing on high-order feature extraction. We can further simplify this into the following form:

$$M_0 = \text{Vecx}(I + \vec{x} + \vec{y}) = \vec{x}A^1, \quad N_0 = \vec{y}(I + \vec{x} + \vec{y}) = \vec{y}A^1, \quad (12)$$

We then define the next terms M_1 and N_1 based on Eq. 12 as:

$$M_1 = \mathcal{M}(M_0, N_0) = \vec{x}A^1(1 + \vec{x} + \vec{y} + 2\vec{x}\vec{y} + \vec{x}^2 + \vec{y}^2) = \vec{x}A^3, \quad (13)$$

$$N_1 = \mathcal{N}(M_0, N_0) = \vec{y}A^1(1 + \vec{x} + \vec{y} + 2\vec{x}\vec{y} + \vec{x}^2 + \vec{y}^2) = \vec{y}A^3. \quad (14)$$

By following this pattern, we generalize the expressions for M_n and N_n as:

$$M_n = \mathcal{M}(M_{n-1}, N_{n-1}) = \vec{x}A^{2^n-1}(1 + (\vec{x} + \vec{y})A^{2^n-1}) = \vec{x}A^{2^{n+1}-1}, \quad (15)$$

$$N_n = \mathcal{N}(M_{n-1}, N_{n-1}) = \vec{y}A^{2^n-1}(1 + (\vec{x} + \vec{y})A^{2^n-1}) = \vec{y}A^{2^{n+1}-1}, \quad (16)$$

$$M_n + N_n = A^{2^{n+1}}. \quad (17)$$

As shown in Eq. 17, if we set $n = \log_2 N - 1$, we can use $M_n + N_n$ to express all constituent terms in Eq. 8. In our method, we implement this progressive derivation process as a multi-layer network. As shown in Figure 3, each layer corresponds to the iteration from M_{n-1} to M_n . The original input to M_0 and N_0 are the encoded image and text features. This network structure allows us to express all

constituent terms in Eq. 8 with only logarithmic growth in the number of terms, significantly reducing the number of parameters and computational complexity while preserving the ability to capture complex multimodal interactions.

In practice, each term is preceded by a corresponding coefficient matrix (omitted in Eq. 8 for clarity). The final expression for M_n (and similarly for N_n) is given in Eq. 18. The full formulation of the proposed multimodal Taylor series network is presented in Eq. 19.

$$M_n = W_\gamma M_{n-1} + (W_{\alpha_1} M_{n-1}) \odot (W_{\alpha_2} N_{n-1}) + (W_{\beta_1} M_{n-1}) \odot (W_{\beta_2} M_{n-1}), \quad (18)$$

$$f(\vec{x}, \vec{y}) \approx M_n + N_n, \quad \text{where } n = \log_2 N - 1, \quad (19)$$

where $W_\gamma, W_{\alpha_1}, W_{\alpha_2}, W_{\beta_1}, W_{\beta_2} \in \mathbb{R}^{d \times d}$ are coefficient matrices, and $\odot$ denotes the Hadamard product. From Eq. 19, we see that this approach preserves both low-order and high-order information from different modalities while effectively capturing cross-modal correlations.

To incorporate semantic meaning into the notations M_n and N_n , in the experiments, we refer to them as the Text-Guided Image Refinement (TIR) feature and the Image-Guided Text Refinement (ITR) feature, respectively. In TIR, the text modality is dominant, while the image serves as a supervisory signal to adjust inter-modal correlations. Conversely, in ITR, the image modality is dominant and influenced by the text.

3.5 Infinity Norm Scaling

To maintain the original structure and properties of the Taylor series, we avoid introducing nonlinear factors such as sigmoid, tanh, or batch normalization for magnitude scaling within the network architecture. However, without any form of scaling on the data flow within the network, this can lead to a magnitude explosion in the network output, as well as gradient explosion during back-propagation. To address this, we propose an infinity norm scaling strategy.

Specifically, for the n -th layer of the network, we first aggregate a batch of $M_n \in \mathbb{R}^d$ and $N_n \in \mathbb{R}^d$ into $M_{batch} \in \mathbb{R}^{batch \times d}$ and $N_{batch} \in \mathbb{R}^{batch \times d}$, respectively, where $batch$ represents the actual batch size. We then apply the L_∞ norm constraint to M_{batch} and N_{batch} , scaling the values in the batch data for each layer to the range of -1 to 1, as shown in the following formula:

$$M'_{batch} = \frac{M_{batch}}{\|M_{batch}\|_\infty}, \quad N'_{batch} = \frac{N_{batch}}{\|N_{batch}\|_\infty}, \quad (20)$$

where $\|\cdot\|_\infty$ denotes the maximum absolute value within the batch. This ensures that the data flow remains stable, preventing magnitude and gradient explosions.

3.6 Low-Order Independent Initialization

In each layer of the proposed network, the parameters $W_{\alpha_1}, W_{\alpha_2}, W_{\beta_1}, W_{\beta_2}$ for the square and interaction terms in Eq. 18 are initialized using Xavier uniform initialization [9]. For the linear term, we initialize its coefficient matrix W_γ with an additional identity matrix, i.e., $I + \text{xavier}_{init}$, allowing it to act as a residual term as well.

It is crucial to choose an initialization method that ensures the square and interaction terms are nearly zero at the beginning of

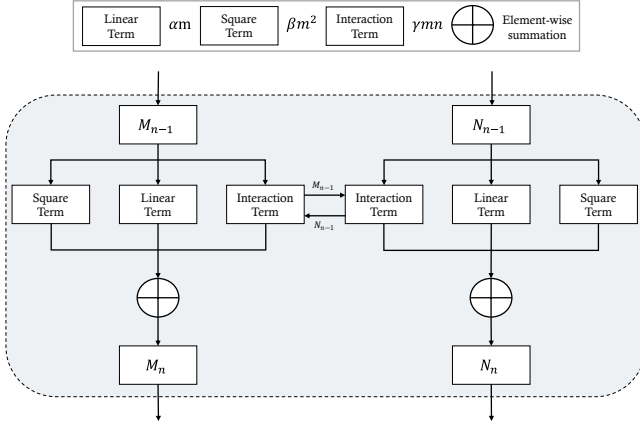


Figure 3: The architecture of an MTS layer. Here, m and n represent the feature vectors of the two modalities in the current layer, while α , β , and γ are learnable parameters.

network training. Otherwise, it would impose a prior assumption on the higher-order features of each modality and the interaction between the two modalities, hindering comprehensive learning. Furthermore, this would not align with the principle illustrated in Figure 1, where the Taylor series progressively fits complex functions by learning from low-order to high-order terms. Next, we briefly demonstrate why our initialization method ensures this.

Assume the feature distribution for the image modality follows $N_{\text{Img}}(0, 1)$ and for the text modality follows $N_{\text{Text}}(0, 1)$, representing two independent normal distributions. After applying the coefficient matrix, the distributions remain unchanged because the Xavier-initialized matrix ensures that the variance of the features before and after input remains consistent.

After applying infinity norm scaling (Section 3.5), the distribution of the interaction term can be expressed as:

$$\text{Inter} = \frac{N_{\text{Img}}(0, 1)}{\mathbb{E}(\text{Max}[N_{\text{Img}}(X_{1,\dots,\text{batch}})])} \cdot \frac{N_{\text{Text}}(0, 1)}{\mathbb{E}(\text{Max}[N_{\text{Text}}(Y_{1,\dots,\text{batch}})])}, \quad (21)$$

where $X_{1,\dots,n}$ and $Y_{1,\dots,n}$ represent a batch of independent samples from the N_{Img} and N_{Text} distributions, respectively, and $\mathbb{E}(\text{Max}[\cdot])$ is the expected value of the L_∞ norm.

From a complex mathematical derivation (details omitted), we know that for samples $X_1, \dots, X_n \sim N(0, 1)$, the approximation of the expected value of the L_∞ norm is:

$$\mathbb{E}(\text{Max}[X_1, \dots, X_n]) \approx \sqrt{2 \log n}. \quad (22)$$

Thus, the distribution function of the interaction term in Eq. 21 can be transformed as follows:

$$\text{InterTerm} \approx \frac{N_{\text{Img, Text}}(0, 1)}{2 \log(\text{batch} \times d)}, \quad (23)$$

where $N_{\text{Img, Text}}(0, 1)$ represents a bivariate normal distribution with a mean of 0 and variance of 1, comprising both image and text features. The expected mean of the distribution in Eq. 23 is 0, and the corresponding variance is $\frac{1}{(2 \log(\text{batch} \times d))^2}$.

Dataset	Weibo	Pheme	Sarcasm
Language	Chinese	English	English
Total Samples	7532	2018	22225
Misinformation	3749	590	9601
Non-Misinformation	3783	1428	12624

Table 1: Statistics of the Datasets.

Given the experimental settings where the batch size is 24 and $d = 768$, the variance is approximately 0.0012, which is very close to 0. This indicates that, after initialization and without any training, the values of the interaction and square terms fluctuate within a narrow range around 0, effectively behaving as if these terms are not included.

In this way, the network retains only the most basic low-order features in the early stages of training. Since the initialization method is applied to all layers, the entire network can initially be viewed as a nearly linear combination of two independent variables, i.e., $f(\vec{x}, \vec{y}) = g(\vec{x}) + h(\vec{y})$. As training progresses, the learning of higher-order coefficients gradually captures the higher-order function distribution. This process also implicitly learns the correlation between the two modalities, achieving modality fusion within the polynomial expression.

4 Experiments

4.1 Experimental Setup

4.1.1 Datasets. We evaluate the proposed method on three benchmark datasets for two tasks: Weibo [24] and Pheme [34] for multimodal fake news detection, and Sarcasm [2] for multimodal sarcasm detection. For Weibo and Sarcasm, we use the partitioning strategy from LogicDM [19], and for Pheme, we follow MFAN's [31] method. Dataset statistics are shown in Table 1. Fake news and sarcasm are treated as misinformation, while real news and non-sarcastic expressions are considered non-misinformation.

4.1.2 Compared Methods. We compare the proposed MTS network with three categories of baselines:

1) Unimodal Models. These methods operate on a single modality: BERT [6] for text, and ViT [7] for images.

2) Multimodal Models Using Only Original Samples. This category includes EANN [25], MVAE [12], SAFE [32], CAFE [4], SpotFake [23], SpotFake+ [22], FSRU [15], and MMCAN [10], which fuse features from the original image and text data without external knowledge, focusing on modality interaction.

3) Multimodal Models Using Additional Information. Models like MCNN [28], MCAN [26], BMR [29], LogicDM [19], and MFAN [31] incorporate additional knowledge or external input data to enhance model context and performance.

The proposed MTS network, unlike the third category, uses only the original image and text data. We include these methods for comparison to demonstrate that even without external knowledge, MTS achieves superior performance, highlighting its strength in capturing multimodal interactions.

Dataset	Methods	Accuracy	Fake News			Real News		
			Precision	Recall	F1-Score	Precision	Recall	F1-Score
Weibo	EANN [25]	78.2	82.7	69.7	75.6	75.2	86.3	80.4
	MVAE [12]	82.4	85.4	76.9	80.9	80.2	87.5	83.7
	SAFE [32]	76.3	83.3	65.9	73.6	71.7	86.8	78.5
	CAFE [4]	84.0	85.5	83.0	84.2	82.5	85.1	83.7
	SpotFake [23]	86.9	87.7	85.9	86.8	86.1	87.9	87.0
	SpotFake+ [22]	87.0	88.7	84.9	86.8	85.5	89.2	87.3
	FSRU [15]	90.1	92.2	89.2	<u>90.6</u>	87.9	91.3	89.5
	MMCAN-Res [10]	<u>90.6</u>	91.6	<u>89.7</u>	<u>90.6</u>	<u>89.8</u>	91.6	<u>90.7</u>
	MCNN* [28]	84.6	80.9	85.7	83.2	87.9	83.7	85.8
	MCAN* [26]	87.8	90.1	86.2	88.1	87.0	90.3	88.6
	BMR* [29]	88.4	87.5	88.6	88.0	87.4	88.1	87.7
	LogicDM* [19]	85.2	86.2	84.5	85.3	84.3	85.9	85.1
PHEME	MFAN* [31]	89.1	94.2	83.5	88.5	85.0	94.8	89.6
	MTS (Ours)	91.9	93.7	90.7	92.2	90.1	93.3	91.6
	EANN [25]	68.1	68.5	66.4	69.4	70.1	75.0	74.7
	MVAE [12]	85.2	80.6	71.9	76.0	89.1	91.7	89.3
	SAFE [32]	81.1	82.7	55.9	66.7	80.6	94.0	86.6
	CAFE [4]	86.1	81.2	64.5	71.9	87.5	<u>94.3</u>	90.8
	SpotFake [23]	82.3	74.3	74.5	74.4	86.4	86.6	86.3
	SpotFake+ [22]	80.0	73.0	66.8	69.7	83.2	86.9	85.0
	MMCAN-Res [10]	<u>89.0</u>	80.3	79.4	79.9	92.2	92.6	<u>92.4</u>
	MCNN* [28]	82.4	80.9	77.9	79.5	83.9	87.0	85.4
	MCAN* [26]	86.7	<u>81.9</u>	<u>82.1</u>	<u>82.0</u>	88.7	88.5	88.6
	MFAN* [31]	88.8	77.1	84.6	80.7	93.9	90.5	92.2
	MTS (Ours)	91.4	89.2	80.5	84.7	<u>92.2</u>	96.0	94.1

Table 2: Performance (%) on the multimodal fake news detection task across the Weibo and PHEME datasets. Methods marked with * incorporate external knowledge or auxiliary information. The best results for each metric are highlighted in bold, and the second-best results are underlined. Some baselines are excluded from the PHEME comparison due to unavailable results.

4.1.3 Implementation Details. For text modality, we use the BERT-base-chinese model¹ for the Chinese Weibo dataset, and the BERT-base-uncased model² for English datasets (PHEME and Sarcasm), with a maximum token limit of 200. For image modality, we employ ResNet34³ as the encoder, resizing input images to 224x224 pixels. The outputs from both modalities are projected into 768-dimensional vectors using a linear layer and fed into the MTS network. The model is trained with cross-entropy loss and optimized using the AdamW optimizer with a learning rate of 2×10^{-5} , and a batch size of 24 for all experiments.

4.1.4 Evaluation Metrics. To evaluate our method and compare it with baseline models, we use standard performance metrics including accuracy (Acc), precision (P), recall (R), and F1 score (F1).

4.2 Comparison Results

Multimodal Fake News Detection. As shown in Table 2, the proposed MTS network consistently delivers superior performance across both the Weibo and PHEME datasets, outperforming all baseline methods on the majority of metrics. Compared to baselines that do not utilize external information, MTS demonstrates significant

¹<https://huggingface.co/bert-base-chinese>

²<https://huggingface.co/bert-base-uncased>

³<https://pytorch.org/vision/main/models/generated/torchvision.models.resnet34>

Model		Acc	P	R	F1
Unimodal	Bert [6]	83.9	78.7	82.3	80.2
	ViT [7]	67.8	57.9	70.1	63.4
Multimodal	HFM [3]	83.4	76.6	84.3	80.2
	D&R Net [27]	84.0	78.0	83.4	80.6
	Att-Bert [21]	86.1	80.9	85.1	82.9
	InCrossMGs [16]	86.1	81.4	84.4	82.8
	HCM [18]	87.4	81.8	<u>86.5</u>	84.1
	LogicDM* [19]	<u>88.1</u>	85.7	85.0	<u>85.3</u>
	MTS (Ours)	89.5	89.0	89.4	89.2

Table 3: Performance (%) on multimodal sarcasm detection.

improvements. For instance, compared to the competitive baseline MMCAN-Res, on the Weibo dataset, MTS improves the F1-Score for fake news and real news by 1.6% and 0.9%, respectively. Similarly, on the PHEME dataset, MTS improves the F1-Score for fake news by 4.8% and for real news by 1.7%.

Even when compared to methods that leverage external knowledge, such as MFAN, MTS still outperforms them across most metrics. For example, MTS improves accuracy by 2.8% on Weibo and by 2.6% on PHEME, further underscoring its effectiveness without relying on additional external data.

Row	Method	Weibo				Sarcasm			
		Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
1	-w/o Inter	91.5	91.6	91.3	91.4	84.6	84.1	84.0	84.0
2	-w/o Square	91.4	91.4	91.3	91.3	89.5	89.1	89.4	89.2
3	-w/o ITR	90.7	90.7	90.7	90.7	87.6	87.0	87.4	87.2
4	-w/o TIR	89.6	90.1	89.3	89.5	89.4	89.0	89.1	89.1
5	MTS	91.5	91.6	91.4	91.5	89.5	89.1	89.2	89.2

Table 4: Ablation studies on the different components of the proposed method. "Inter" denotes the interaction term. "Square" denotes the square term.

Multimodal Sarcasm Detection. As shown in Table 3, the MTS network significantly outperforms all baseline methods in the sarcasm detection task. Notably, compared to the competitive LogicDM model, MTS improves accuracy by 1.4% and F1-Score by 3.9%, illustrating the model's capacity for handling nuanced multimodal content.

These results strongly demonstrate the effectiveness of the proposed MTS network in capturing deep interactions between different modalities. The consistent improvements across both fake news and sarcasm detection tasks highlight the model's versatility, making it a promising solution for a wide range of multimodal classification problems.

4.3 Ablation Study

As shown in Table 4, the ablation study investigates the contributions of different components in the proposed model. First, comparing Row 5 with Row 1, we observe a significant performance drop on the Sarcasm dataset when the interaction term is removed. This highlights the importance of the interaction term in capturing cross-modal correlations between text and image. Second, comparing Row 5 with Row 2, we find that removing the square term has minimal impact, likely because high-order polynomial information, which the square term captures, can also be learned through the interaction term. Lastly, comparing Row 5 with Rows 3 and 4, we see that the absence of either the ITR or TIR feature leads to reduced performance, demonstrating the necessity of both features in the proposed network for optimal performance.

4.4 Analysis of Feature Space Distribution

As shown in Figure 5 (a), we observe a clear trend of decreasing variance as the number of layers increases. Variance in a feature vector typically reflects data dispersion, with higher variance indicating more scattered data points. Initially, the features are dispersed in a high-dimensional space, but as the MTS network deepens and becomes more complex, the features gradually concentrate into a lower-dimensional subspace, eventually converging into a highly-compressed representation for final classification. This behavior is consistent with deep learning models, where features are progressively refined and information becomes more concentrated with each layer.

Qualitative analysis of Weibo feature distributions (Figure 4) demonstrates enhanced class separability: MTS-processed TIR/ITR features (column 2) exhibit clearer boundaries than original features (column 1), directly linking improved classification to cross-modal

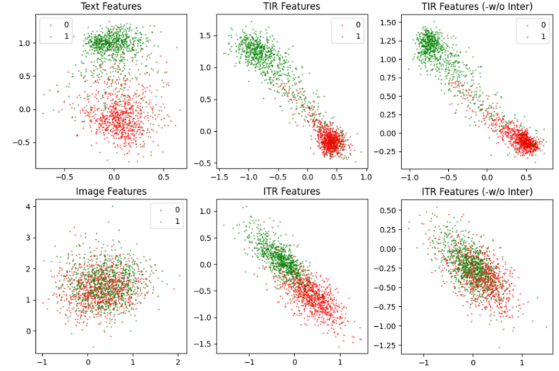


Figure 4: Feature visualization on a two-dimensional plane. Red and green points represent samples from two different classes. "Image Features" and "Text Features" refer to the original image and text features, e_i and e_t , before being input into the MTS network. "ITR Features" and "TIR Features" represent the corresponding output features from the MTS network. The notation "-w/o Inter" indicates that the interaction term has been removed from the features.

Used Layers	Accuracy	Precision	Recall	F1-Score
None	62.0	62.4	62.3	61.9
2	88.5	88.5	88.5	88.5
4	91.6	91.7	91.5	91.5
6	91.4	91.5	91.3	91.4
All	91.5	91.6	91.4	91.5

Table 5: Performance (%) of an 8-layer MTS network on the Weibo dataset using different number of subsets of layers for inference: "None" (no layers), '2', '4', '6', and 'All' (all layers).

refinement. Crucially, interaction term removal in ITR features collapses separation, confirming their necessity for modeling cross-modal correlations, while TIR features remain minimally affected by such ablation – suggesting the text modality's inherent discriminative power in Weibo overshadows interaction effects.

4.5 Model Interpretability from the Perspective of Taylor Series

As shown in Figure 5 (b), we observe that the magnitude of feature changes is substantial in the initial layers but diminishes as the

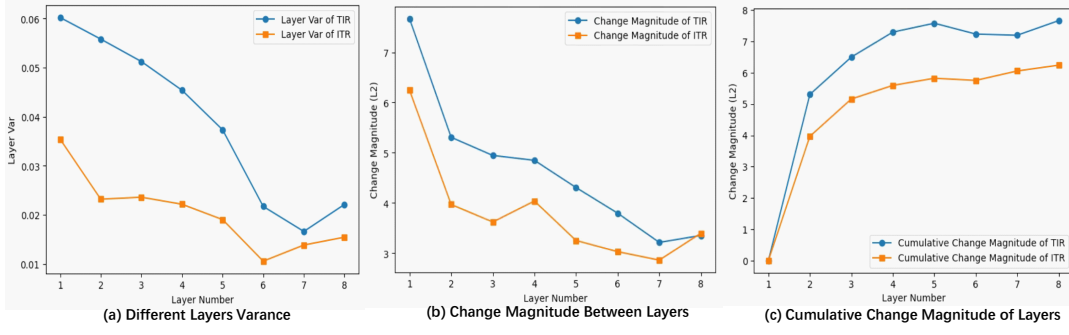


Figure 5: Change curves of various average values across all samples in the Weibo test set: (a) shows the variance of the ITR and TIR features at each layer of the MTS network; (b) depicts the L2 norm difference of features between consecutive layers, where the vertical value at l_n represents the L2 norm difference between features at layers l_n and l_{n-1} ; (c) illustrates the cumulative L2 norm change of features from the first layer to the current layer, where the value at l_n represents the cumulative L2 norm difference between features at layer l_n and the first layer.

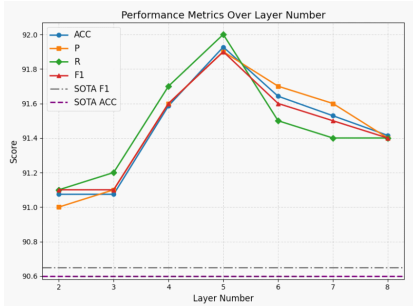


Figure 6: Performance (%) trend on the Weibo dataset with different layer numbers in the MTS network. The horizontal axis shows the number of layers in the network, and the vertical axis represents the performance metrics. For comparison, we also include the Macro F1-Score and accuracy from the competitive baseline MMCAN-Res, labeled as "SOTA F1" and "SOTA ACC."

model deepens. In Figure 5 (c), the cumulative change increases rapidly at first before stabilizing. These patterns align with the theory of Taylor series expansion, where lower-order terms dominate the capture of core feature interactions, while higher-order terms have diminishing influence and serve primarily as fine-tuners for refining the complex function representation.

To further support this view, we train an 8-layer MTS network and evaluate the impact of using different numbers of layers during testing (see Table 5). The results show that performance improves progressively with the first four layers, while the latter four layers contribute minimally to further enhancement. This observation is also consistent with the Taylor series expansion theory, where lower-order terms are more critical for the model's success, and higher-order terms provide finer adjustments to the overall representation.

4.6 Analysis of Layer Number Selection

Evaluations on Weibo (Figure 6) reveal that the MTS network surpasses SOTA baselines in Accuracy/F1-Score across different layer configurations, peaking at 5 layers while underperforming with fewer (2) or excessive (8) layers. This demonstrates balanced layer selection is crucial to avoid under/over-parameterization.

Performance trends show improved results as layers increase from 2 to 5, indicating Taylor series' high-order terms enhance expressiveness to model complex features and deeper cross-modal interactions. Beyond 5 layers, performance drops due to overfitting from excessive high-order terms that reduce generalization capability.

5 Conclusion

In this paper, we propose a novel multimodal Taylor series network for multimodal misinformation detection. Our approach utilizes the polynomial modeling capabilities of the Taylor series to effectively capture the intricate interactions between text and image modalities, while ensuring high interpretability. Additionally, through carefully designed network architecture, we address the challenge of exponential parameter growth and enhance model scalability. Extensive experiments on multiple datasets demonstrate that our method outperforms existing state-of-the-art approaches in both fake news detection and sarcasm detection tasks, even surpassing methods that utilize auxiliary information. Furthermore, our analysis of the different-layer features validate the model's interpretability, while the layer number analysis reveal consistent superiority and robustness. In the future, we hope to extend the proposed method to broader applications in multimodal tasks.

Acknowledgements

This research is supported by the National Natural Science Foundation of China (Grant No. 62302245, 62372252, 62172237, 62077031, 62176028), the Ministry of Education of the People's Republic of China Humanities and Social Sciences Youth Foundation (Grant No. 23YJCZH240), and the NSFC-General Technology Joint Fund for Basic Research (Grant No. U1936105, U1936206).

References

- [1] Sara Abdali, Bhaskar Krishnamachari, et al. 2022. Multi-modal misinformation detection: Approaches, challenges and opportunities. *arXiv preprint arXiv:2203.13883* (2022).
- [2] Yitao Cai, Huiyu Cai, and Xiaojun Wan. 2019. Multi-modal sarcasm detection in twitter with hierarchical fusion model. In *Proceedings of the 57th annual meeting of the association for computational linguistics*. 2506–2515.
- [3] Yitao Cai, Huiyu Cai, and Xiaojun Wan. 2019. Multi-modal sarcasm detection in twitter with hierarchical fusion model. In *Proceedings of the 57th annual meeting of the association for computational linguistics*. 2506–2515.
- [4] Yixuan Chen, Dongsheng Li, Peng Zhang, Jie Sui, Qin Lv, Lu Tun, and Li Shang. 2022. Cross-modal ambiguity learning for multimodal fake news detection. In *Proceedings of the ACM web conference 2022*. 2897–2905.
- [5] Grigorios G Chrysos, Stylianos Moschoglou, Giorgos Bouritsas, Jiankang Deng, Yannis Panagakis, and Stefanos Zafeiriou. 2021. Deep polynomial neural networks. *IEEE transactions on pattern analysis and machine intelligence* 44, 8 (2021), 4021–4034.
- [6] Jacob Devlin. 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805* (2018).
- [7] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xi-aohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. 2020. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929* (2020).
- [8] Viet Duong, Qiong Wu, Zhengyi Zhou, Hongjue Zhao, Chenxiang Luo, Eric Zavesky, Huaxiu Yao, and Huajie Shao. 2024. CAT: Interpretable Concept-based Taylor Additive Models. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 723–734.
- [9] Xavier Glorot and Yoshua Bengio. 2010. Understanding the difficulty of training deep feedforward neural networks. In *Proceedings of the thirteenth international conference on artificial intelligence and statistics*. JMLR Workshop and Conference Proceedings, 249–256.
- [10] Linmei Hu, Ziwang Zhao, Weijian Qi, Xuemeng Song, and Liqiang Nie. 2024. Multimodal matching-aware co-attention networks with mutual knowledge distillation for fake news detection. *Information Sciences* 664 (2024), 120310.
- [11] Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, et al. 2023. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *arXiv preprint arXiv:2311.05232* (2023).
- [12] Dhruv Khattar, Jaipal Singh Goud, Manish Gupta, and Vasudeva Varma. 2019. Mvae: Multimodal variational autoencoder for fake news detection. In *The world wide web conference*. 2915–2921.
- [13] Tamara G Kolda and Brett W Bader. 2009. Tensor decompositions and applications. *SIAM review* 51, 3 (2009), 455–500.
- [14] An Lao, Chongyang Shi, and Yayi Yang. 2021. Rumor detection with field of linear and non-linear propagation. In *Proceedings of the Web Conference 2021*. 3178–3187.
- [15] An Lao, Qi Zhang, Chongyang Shi, Longbing Cao, Kun Yi, Liang Hu, and Duoqian Miao. 2024. Frequency spectrum is more effective for multimodal representation and fusion: A multimodal spectrum rumor detector. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 18426–18434.
- [16] Bin Liang, Chenwei Lou, Xiang Li, Lin Gui, Min Yang, and Ruifeng Xu. 2021. Multi-modal sarcasm detection with interactive in-modal and cross-modal graphs. In *Proceedings of the 29th ACM international conference on multimedia*. 4707–4715.
- [17] Victor Weixin Liang, Yuhui Zhang, Yongchan Kwon, Serena Yeung, and James Y Zou. 2022. Mind the gap: Understanding the modality gap in multi-modal contrastive representation learning. *Advances in Neural Information Processing Systems* 35 (2022), 17612–17625.
- [18] Hui Liu, Wenya Wang, and Haoliang Li. 2022. Towards multi-modal sarcasm detection via hierarchical congruity modeling with knowledge enhancement. *arXiv preprint arXiv:2210.03501* (2022).
- [19] Hui Liu, Wenya Wang, and Haoliang Li. 2023. Interpretable multimodal misinformation detection with logic reasoning. *arXiv preprint arXiv:2305.05964* (2023).
- [20] Pablo Morala, Jenny Alexandra Cifuentes, Rosa E Lillo, and Iñaki Ucar. 2023. NN2Poly: A polynomial representation for deep feed-forward artificial neural networks. *IEEE Transactions on Neural Networks and Learning Systems* (2023).
- [21] Hongliang Pan, Zheng Lin, Peng Fu, Yatao Qi, and Weiping Wang. 2020. Modeling intra and inter-modality incongruity for multi-modal sarcasm detection. In *Findings of the Association for Computational Linguistics: EMNLP 2020*. 1383–1392.
- [22] Shivangi Singhal, Anubha Kabra, Mohit Sharma, Rajiv Ratn Shah, Tanmoy Chakraborty, and Ponnurangam Kumaraguru. 2020. Spotfake+: A multimodal framework for fake news detection via transfer learning (student abstract). In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 34. 13915–13916.
- [23] Shivangi Singhal, Rajiv Ratn Shah, Tanmoy Chakraborty, Ponnurangam Kumaraguru, and Shin'ichi Satoh. 2019. Spotfake: A multi-modal framework for fake news detection. In *2019 IEEE fifth international conference on multimedia big data (BigMM)*. IEEE, 39–47.
- [24] Changhe Song, Cheng Yang, Huimin Chen, Cunchao Tu, Zhiyuan Liu, and Maosong Sun. 2019. CED: Credible early detection of social media rumors. *IEEE Transactions on Knowledge and Data Engineering* 33, 8 (2019), 3035–3047.
- [25] Yaqing Wang, Fenglong Ma, Zhiwei Jin, Ye Yuan, Guangxu Xun, Kishlay Jha, Lu Su, and Jing Gao. 2018. Eann: Event adversarial neural networks for multi-modal fake news detection. In *Proceedings of the 24th acm sigkdd international conference on knowledge discovery & data mining*. 849–857.
- [26] Yang Wu, Pengwei Zhan, Yunjian Zhang, Liming Wang, and Zhen Xu. 2021. Multimodal fusion with co-attention networks for fake news detection. In *Findings of the association for computational linguistics: ACL-IJCNLP 2021*. 2560–2569.
- [27] Nan Xu, Zhixiong Zeng, and Wenji Mao. 2020. Reasoning with multimodal sarcastic tweets via modeling cross-modality contrast and semantic association. In *Proceedings of the 58th annual meeting of the association for computational linguistics*. 3777–3786.
- [28] Junxiao Xue, Yabo Wang, Yichen Tian, Yafei Li, Lei Shi, and Lin Wei. 2021. Detecting fake news by exploring the consistency of multimodal data. *Information Processing & Management* 58, 5 (2021), 102610.
- [29] Qichao Ying, Xiaoxiao Hu, Yangming Zhou, Zhenxing Qian, Dan Zeng, and Shiming Ge. 2023. Bootstrapping multi-view representations for fake news detection. In *Proceedings of the AAAI conference on Artificial Intelligence*, Vol. 37. 5384–5392.
- [30] Xueyao Zhang, Juan Cao, Xirong Li, Qiang Sheng, Lei Zhong, and Kai Shu. 2021. Mining dual emotion for fake news detection. In *Proceedings of the web conference 2021*. 3465–3476.
- [31] Jiaqi Zheng, Xi Zhang, Sanchuan Guo, Quan Wang, Wenyu Zang, and Yongdong Zhang. 2022. MFAN: Multi-modal Feature-enhanced Attention Networks for Rumor Detection. In *IJCAI*, Vol. 2022. 2413–2419.
- [32] Xinyi Zhou, Jindi Wu, and Reza Zafarani. 2020. Similarity-aware multi-modal fake news detection. In *Pacific-Asia Conference on knowledge discovery and data mining*. Springer, 354–367.
- [33] Yangming Zhou, Yuzhou Yang, Qichao Ying, Zhenxing Qian, and Xinpeng Zhang. 2023. Multimodal fake news detection via clip-guided learning. In *2023 IEEE International Conference on Multimedia and Expo (ICME)*. IEEE, 2825–2830.
- [34] Arkaitz Zubiega, Maria Liakata, and Rob Procter. 2017. Exploiting context for rumour detection in social media. In *Social Informatics: 9th International Conference, SocInfo 2017, Oxford, UK, September 13-15, 2017, Proceedings, Part I 9*. Springer, 109–123.